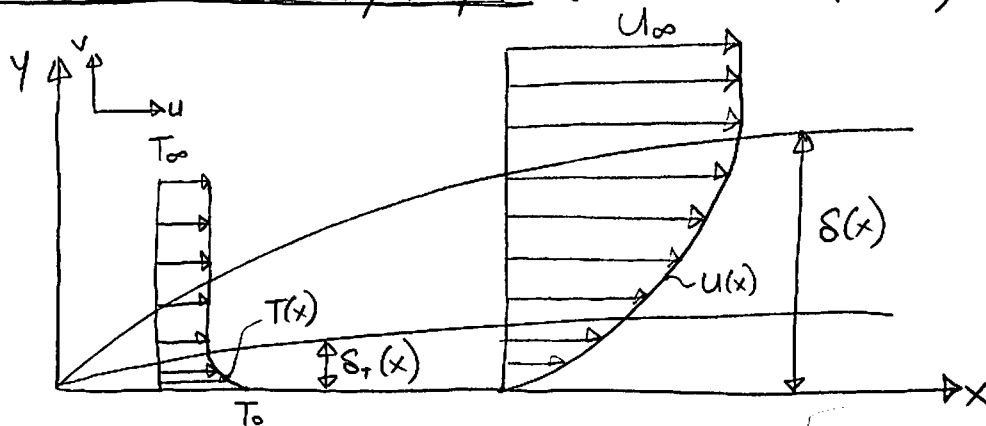


Thin Thermal Boundary Layer ($\delta \approx \delta_T$ or $\delta_T \ll \delta$)



In this limit, the thermal boundary layer is much thinner than the hydrodynamic boundary layer.

Here it gets a bit tricky. When scaling u in the energy equation, we must use the u in the thermal boundary layer, which in this case is not U_∞ !

$$u \neq U_\infty$$

$$u \sim U_\infty \left(\frac{\delta_T}{\delta} \right) \quad (\text{Linear approximation} \Rightarrow \text{ok for scaling})$$

Now we can deal with v :

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \Rightarrow \frac{u}{L} + \frac{v}{\delta_T} = 0 \Rightarrow \text{Note here I used } \delta_T \text{ instead of } \delta. \text{ Again, this is due to } \delta_T \text{ being well within } \delta \text{ and the length scale being pertinent to heat trans.}$$

$$v \sim u \left(\frac{\delta_T}{L} \right) \sim U_\infty \frac{\delta_T^2}{\delta L}$$

Now we can check our convective terms again to see the dominant ones

$$\left. \begin{aligned} u \frac{\Delta T}{L} &\sim U_\infty \frac{\delta_T}{\delta} \cdot \frac{\Delta T}{L} \\ v \frac{\Delta T}{\delta_T} &\sim U_\infty \frac{\delta_T^2}{\delta L} \cdot \frac{\Delta T}{\delta_T} \sim U_\infty \frac{\delta_T}{\delta} \frac{\Delta T}{L} \end{aligned} \right\} \text{Both are important \& must be kept}$$

Back into our energy equation:
convection $\sim$ conduction

$$U_{\infty} \frac{\delta_T}{\delta} \frac{\Delta T}{L} \sim \alpha \frac{\Delta T}{\delta_T^2}$$

We know from our hydrodynamic solution that: $\frac{\delta}{L} \sim Re_L^{-1/2}$
Back substituting; (inertia $\sim$ friction)

$$U_{\infty} \frac{\delta_T}{L Re_L^{-1/2}} \cdot \frac{1}{L} \sim \alpha \frac{1}{\delta_T^2} \quad ; \quad Pr = \frac{U}{\alpha} \Rightarrow \alpha = \frac{U}{Pr}$$

$$U_{\infty} \frac{\delta_T^3}{L^2 Re_L^{-1/2}} \sim \frac{U}{Pr}$$

$$\frac{\delta_T^3}{L^3} \sim \frac{U Re_L^{-1/2}}{Pr \cdot L U_{\infty}} \sim \underbrace{\left(\frac{U}{U_{\infty} L} \right)}_{Re_L^{-2}} Pr^{-1} Re_L^{-1/2}$$

$$\frac{\delta_T}{L} \sim \left(Pr^{-1} Re_L^{-3/2} \right)^{1/3}$$

$$\boxed{\frac{\delta_T}{L} \sim Pr^{-1/3} Re_L^{-1/2}} \quad \text{for } \frac{\delta_T}{\delta} \ll 1.$$

This means that:

$$\frac{\delta_T}{\delta} = \left(\frac{\delta_T}{L} \right) \cdot \left(\frac{L}{\delta} \right) \sim Pr^{-1/3} Re_L^{-1/2} \cdot Re_L^{1/2}$$

$$\boxed{\frac{\delta_T}{\delta} \sim Pr^{-1/3} \ll 1} \Rightarrow \text{This is valid for fluids with } Pr^{1/3} \gg 1.$$

These include highly viscous fluids such as oils, honey, etc..
Solving for h & Nu :

$$\boxed{h \sim \frac{k}{\delta_T} \sim \frac{k}{L} Pr^{1/3} Re_L^{1/2}} \quad (Pr \gg 1)$$

$$\boxed{Nu = \frac{hL}{k} \sim Pr^{1/3} Re_L^{1/2}} \quad (Pr \gg 1)$$

Note we could have done the scaling in terms of a timescale analysis:

$$\text{Conduction} \Rightarrow \tau_{sT} \sim \frac{\delta_T^2}{\alpha} \quad (\text{from } 0 \sim \sqrt{\alpha t})$$

$$\text{Convection} \Rightarrow \tau_L \sim \frac{L}{U_\infty} \quad (\text{for } Pr < 1) \quad \text{Longitudinal speed} \sim U_\infty$$

$$\tau_L \sim \frac{L}{\frac{\delta_T}{s} U_\infty} \quad (\text{for } Pr > 1) \quad \text{Longitudinal speed} \sim \frac{\delta_T}{s} U_\infty$$

Scaling our timescales with each other:

Pr < 1:

$$\tau_{sT} \sim \tau_L \Rightarrow \frac{\delta_T^2}{\alpha} \sim \frac{L}{U_\infty} \Rightarrow Pe = \frac{U_\infty L}{\alpha}$$

$$\boxed{\delta_T \sim \sqrt{\frac{\alpha L}{U_\infty}} \sim L \cdot Pe^{-1/2}} \Rightarrow \text{Same as before.}$$

Pr > 1:

$$\tau_{sT} \sim \tau_L \Rightarrow \frac{\delta_T^2}{\alpha} \sim \frac{L}{\frac{\delta_T}{s} U_\infty}$$

$$\delta_T^3 \sim \frac{s L \alpha}{U_\infty} \Rightarrow \delta \sim L \cdot Re_L^{-1/2}$$

$$\delta_T^3 \sim \frac{L^2 \alpha Re_L^{-1/2}}{U_\infty} \sim \frac{L^2 \nu}{U_\infty Pr} \cdot Re_L^{-1/2}$$

$$\boxed{\delta_T \sim L \cdot Pr^{-1/3} Re_L^{-1/2}} \Rightarrow \text{Same as before.}$$

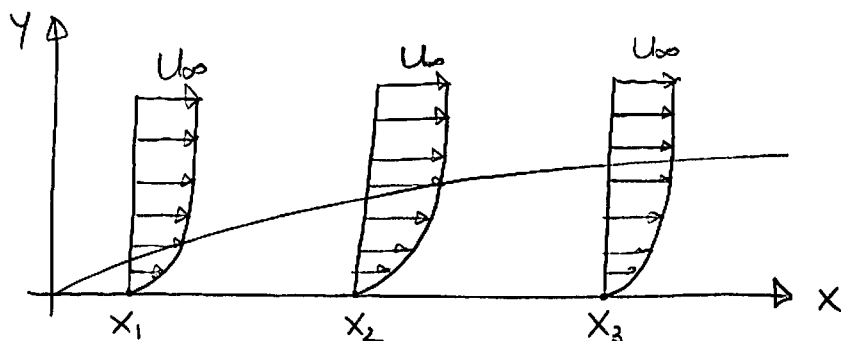
Meaning of Reynolds Number

For external flows, $Re_L = U_\infty L / \nu$. For all of your lives you've been told that $Re_L = \text{Inertia} / \text{Friction}$. But we just said that for laminar b.l. $\text{Inertia} \sim \text{Friction}$, up to $10^5 = Re_L$. The way to interpret Reynolds number in b.l. flow is geometric:

$$\boxed{Re_L^{1/2} = \frac{L}{\delta} = \frac{\text{wall length}}{\text{b.l. thickness}}} \Rightarrow \text{sometimes called "slenderness" ratio}$$

Similarity Solutions

The basic idea to these solutions is the observation that from one location x to another, the U & T profiles look similar. For example:



For all three locations, we know that $U(x=0) = 0$
 $U(x \rightarrow \infty) = U_\infty$

From this, we can argue that:

$$\frac{U}{U_\infty} = f'(\eta) \Rightarrow \eta \equiv \text{similarity variable.}$$

We can intuitively say that $\eta \sim y$, and $\eta = f(x)$.
 From our previous scaling analysis, we can deduce:

$$\eta \sim \frac{y}{\delta(x)} \sim \frac{y}{x Re_x^{-1/2}}$$

So we can assume:

$$\boxed{\eta = \frac{y}{x} Re_x^{1/2} = y \sqrt{\frac{U_\infty}{\nu x}}} \Rightarrow \text{Similarity Variable.}$$

We also assume that:

$$\boxed{\frac{U}{U_\infty} = f'(\eta)}$$

Our equations to work with are:

$$U \frac{\partial U}{\partial x} + v \frac{\partial U}{\partial y} = \nu \frac{\partial^2 U}{\partial y^2}$$

$$\frac{\partial U}{\partial x} + \frac{\partial v}{\partial y} = 0$$

Back substituting and solving for individual terms:

$$\frac{\partial u}{\partial x} = U_{\infty} \frac{\partial f''}{\partial \eta} \cdot \frac{\partial \eta}{\partial x} = U_{\infty} f'' \frac{\partial \eta}{\partial x}$$

$$\frac{\partial \eta}{\partial x} = \frac{\partial}{\partial x} \left(\gamma \sqrt{\frac{U_{\infty}}{\nu}} \cdot x^{-1/2} \right) = \gamma \sqrt{\frac{U_{\infty}}{\nu}} \cdot -\frac{1}{2} x^{-3/2} = -\frac{1}{2x} \eta$$

$$\frac{\partial u}{\partial x} = -U_{\infty} f'' \frac{1}{2x} \eta$$

Now we can use continuity to solve for v .

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

$$\frac{\partial v}{\partial y} = U_{\infty} f'' \frac{1}{2x} \eta$$

$$\frac{v}{U_{\infty}} = \frac{1}{2x} \int_0^y f'' \eta dy = \underbrace{\frac{1}{2x} \sqrt{\frac{xU_{\infty}}{\nu}}}_{\frac{1}{2} Re_x^{-1/2}} \int_0^{\eta} f'' \eta d\eta \quad (\text{since } dy = \sqrt{\frac{xU_{\infty}}{\nu}} d\eta)$$

Now we can solve the above integral

$$\int_0^{\eta} \eta f'' d\eta = uv - \int v du \quad (\text{IBP: } \int u dv = uv - \int v du)$$

$$\text{here } u = \eta, \quad v = f', \quad dv = f'' d\eta$$

$$\begin{aligned} \int_0^{\eta} \eta f'' d\eta &= \eta f' - \int_0^{\eta} f' d\eta \\ &= \eta f' - f \end{aligned}$$

$$\text{So: } \frac{v}{U_{\infty}} = \frac{1}{2} Re_x^{-1/2} (\eta f' - f)$$

Now we can solve for our last term:

$$\frac{\partial^2 u}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial u}{\partial y} \right) = \frac{\partial}{\partial y} \left(U_{\infty} \frac{\partial f''}{\partial \eta} \cdot \frac{\partial \eta}{\partial y} \right) = \frac{\partial}{\partial y} \left(U_{\infty} f'' \frac{Re_x^{1/2}}{x} \right)$$

$$= U_{\infty} \frac{Re_x^{1/2}}{x} \frac{\partial}{\partial y} (f'') = \frac{U_{\infty} Re_x^{1/2}}{x} \frac{\partial f''}{\partial \eta} \cdot \frac{\partial \eta}{\partial y}$$

$$\frac{\partial^2 u}{\partial y^2} = \frac{U_{\infty} Re_x}{x^2} f'''$$

Back substituting all of our terms into our momentum eqn.

$$\cancel{U_{\infty} f'} \left(-U_{\infty} f'' \frac{1}{2x} \eta \right) + \cancel{U_{\infty}} \frac{1}{2} Re_x^{-1/2} (\eta f' - f) \left(U_{\infty} f'' \frac{Re_x^{1/2}}{x} \right) \\ = \cancel{U} \frac{U_{\infty} Re_x}{x^2} f'''$$

$$-f' f'' \cancel{U_{\infty}} \frac{\eta}{2} + \frac{1}{2} Re_x^{-1/2} \eta f' \cancel{U_{\infty}} f'' Re_x^{1/2} - \frac{1}{2} Re_x^{-1/2} f \cancel{U_{\infty}} f'' Re_x^{1/2} \\ - \cancel{U} \frac{Re_x}{x} f''' = 0$$

$$-\frac{1}{2} f U_{\infty} f'' - \cancel{U} \frac{Re_x}{x} f''' = 0$$

We know $Re_x = \frac{U_{\infty} x}{\nu}$, back substituting

$$\frac{1}{2} f \cancel{U_{\infty}} f'' + \cancel{U} \frac{U_{\infty} x}{x \cancel{\nu}} f''' = 0$$

$$\boxed{\frac{1}{2} f f'' + f''' = 0}$$

$$\boxed{f'(\eta) = \frac{u}{U_{\infty}}}$$

$$\boxed{\eta = y \sqrt{\frac{U_{\infty}}{\nu x}}}$$

Now we've converted our PDE into an ODE
Converting our boundary conditions:

$$u(y=0) = 0 \Rightarrow \eta(y=0) = 0 ; f'(0) = 0$$

$$u(y \rightarrow \infty) = U_{\infty} \Rightarrow \eta(y \rightarrow \infty) = \infty ; f'(\eta \rightarrow \infty) = 1$$

$$v(y=0) = 0 \Rightarrow \eta(y=0) = 0 ; v = U_{\infty} \frac{1}{2} Re_x^{-1/2} (\eta(0) \cdot f'(0) - f(0)) \\ f(0) = 0$$

To solve, we can assume an infinite series solution:

$$\begin{aligned} f &= a_0 + a_1 \eta + a_2 \eta^2 + a_3 \eta^3 + \dots + \\ f' &= a_1 + 2a_2 \eta + 3a_3 \eta^2 + \dots + \\ f'' &= 2a_2 + 6a_3 \eta + \dots + \\ f''' &= 6a_3 + \dots + \end{aligned}$$

Back substituting and solving for our coefficients (matrix form)

$$\underbrace{(\dots)}_0 \eta^0 + \underbrace{(\dots)}_0 \eta^1 + \underbrace{(\dots)}_0 \eta^2 + \underbrace{(\dots)}_0 \eta^3 + \dots + \underbrace{(\dots)}_0 \eta^n = 0$$

We can obtain a recursion formula relating our constants

$$f = \frac{a_2 \eta^2}{2!} - \frac{a_2 \eta^5}{2 \cdot 5!} + \frac{11}{4} \frac{a_3^3 \eta^8}{8!} + \dots$$

$\Rightarrow$ Blasius Solution (1911)
Prandtl's PhD Student

$$a_2 = 0.332 \Rightarrow \text{since } f(0) = f'(0) = 0 \Rightarrow a_1 = a_0 = 0$$

We know that $f'(\eta) = \frac{U}{U_\infty}$

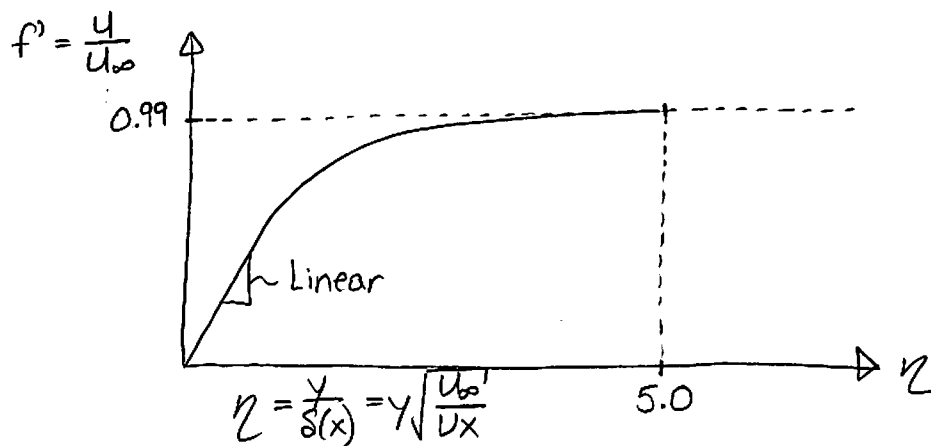
$$f'(\eta) = \frac{2a_2 \eta}{2} + \frac{5a_2 \eta^4}{2 \cdot 120} + \left(\frac{11}{4}\right) \frac{8a_3^3 \eta^7}{40320} + \dots$$

$$\eta = \gamma \sqrt{\frac{U_\infty}{Ux}}$$

Looking at only our first term: $\eta \rightarrow 0$ (higher order terms drop)

$$f'(\eta) = \frac{U}{U_\infty} = 0.332 \gamma \sqrt{\frac{U_\infty}{Ux}} \quad \text{or} \quad \boxed{U(x, \gamma) = 0.332 \gamma \sqrt{\frac{U_\infty^3}{Ux}}}$$

Note we can solve numerically and plot our non-dimensional result. Also, the above $U(x, \gamma)$ result is only valid near the wall where $\eta \ll 1$, so h.o.T drop out.



Solving for when $f' = 0.99$ (when $u = 0.99 u_\infty = \text{b.l. thickness}$)

$$5.0 = \delta \sqrt{\frac{u_\infty}{\nu x}} = \frac{\delta}{x} Re_x^{1/2}$$

$$\boxed{\delta = \frac{5x}{\sqrt{Re_x}}}$$

$\Rightarrow$ Hydrodynamic boundary layer thickness on a flat plate in laminar flow.

Now we can use this super usefull information to do some calculations:

$$\tau(x) = \mu \left. \frac{\partial u}{\partial y} \right|_{y=0} \Rightarrow \text{we know } f' = \overset{a_2}{0.332} \gamma \sqrt{\frac{u_\infty}{\nu x}} = \frac{u}{u_\infty} \text{ at } \eta \rightarrow 0$$

$$u = 0.332 u_\infty \gamma \sqrt{\frac{u_\infty}{\nu x}} \Rightarrow \text{only the first term from our solution since } \eta \ll 1, \text{ and higher order terms drop out.}$$

$$\left. \frac{\partial u}{\partial y} \right|_{y=0} = 0.332 u_\infty \sqrt{\frac{u_\infty}{\nu x}}$$

$$\begin{aligned} \tau(x) &= 0.332 u_\infty \mu \sqrt{\frac{u_\infty \rho}{\mu x}} = 0.332 u_\infty \sqrt{\frac{\mu u_\infty \rho}{x}} \cdot \left(\frac{\rho u_\infty}{\rho u_\infty} \right) \left(\frac{2}{2} \right) \\ &= 0.664 \cdot \frac{1}{2} \cdot \rho u_\infty^2 \sqrt{\frac{\mu u_\infty \rho}{\rho^2 u_\infty^2 x}} = 0.664 \cdot \frac{1}{2} \cdot \rho u_\infty^2 \cdot Re_x^{-1/2} \end{aligned}$$

$$\boxed{\frac{\tau(x)}{\frac{1}{2} \rho u_\infty^2} = 0.664 \cdot Re_x^{-1/2} = C_{f,x}} \Rightarrow \text{Skin friction coefficient}$$

Typically, we want the averaged friction coefficient:

$$\bar{\tau} = \frac{1}{L} \int_0^L \tau(x) dx$$

Noting that $\tau(x) = C x^{-1/2}$, $C = \frac{1}{2} \rho U_\infty^2 \cdot (0.664) U_\infty^{1/2} L^{-1/2}$

$$\bar{\tau} = C \cdot \frac{1}{L} \int_0^L \frac{dx}{x^{1/2}} = \frac{2C}{L^{1/2}}$$

$$\boxed{\bar{\tau} = 0.664 \rho U_\infty^2 \cdot Re_L^{-1/2}} \Rightarrow \text{Wall averaged shear}$$

$$\boxed{\bar{C}_f = \frac{\bar{\tau}}{\frac{1}{2} \rho U_\infty^2} = \frac{1.328}{Re_L^{1/2}}} \Rightarrow \text{Wall averaged skin friction coefficient.}$$

Note on page 41 of the notes, we solved from similarity:

$$\left. \begin{array}{l} \bar{\tau} \sim \rho U_\infty^2 Re_L^{-1/2} \\ \bar{C}_f \sim Re_L^{-1/2} \end{array} \right\} \text{Pretty good when considering the ease in which we got these.}$$

Example Calculate the boundary layer thickness at the windshield of a car moving at 70 mph.

$$U_\infty = 70 \text{ mph } (31.11 \text{ m/s})$$

$$\nu_{\text{air}} = 1.5 \times 10^{-5} \text{ m}^2/\text{s}$$

$$L = 1 \text{ m (length from the front bumper to the windshield)}$$

$$Re_L = \frac{U_\infty L}{\nu} = \frac{(31.11 \text{ m/s})(1 \text{ m})}{(1.5 \times 10^{-5} \text{ m}^2/\text{s})} \approx 2.07 \times 10^6$$

$$\delta \approx \frac{5L}{\sqrt{Re_L}} = \frac{5 \text{ m}}{\sqrt{2.07 \times 10^6}} = 0.0035 \text{ m} = 3.5 \text{ mm}$$

$$\boxed{\delta = 3.5 \text{ mm}} \Rightarrow \text{Very thin!}$$

No wonder Prandtl had a hard time seeing it!